Symptom List:

P0135-O2 SENSOR 1/1 HEATER PERFORMANCE
P0141-O2 SENSOR 1/2 HEATER PERFROMANCE
P0155-O2 SENSOR 2/1 HEATER PERFORMANCE
P0161-O2 SENSOR 2/2 HEATER PERFORMANCE

Test Note: All symptoms listed above are diagnosed using the same tests. The title for the tests will be P0135-O2 SENSOR 1/1 HEATER PERFORMANCE.

When Monitored and Set Condition:

P0135-O2 SENSOR 1/1 HEATER PERFORMANCE

When Monitored: Engine running and heater duty cycle greater than 0%. ASD Relay is energized. Battery voltage greater than 10.4 volts.

Set Condition: No sensor output is received when the PCM powers up the sensor heater. O2 heater is out of control for 128 seconds after it has reached 300 deg C. Two trip fault

P0141-O2 SENSOR 1/2 HEATER PERFROMANCE

When Monitored: Engine running and heater duty cycle greater than 0%. ASD Relay is energized. Battery voltage greater than 10.4 volts.

Set Condition: No sensor output is received when the PCM powers up the sensor heater. O2 heater is out of control for 128 seconds after it has reached 350 deg C. Two trip fault.

P0155-O2 SENSOR 2/1 HEATER PERFORMANCE

When Monitored: Engine running and heater duty cycle greater than 0%. ASD Relay is energized. Battery voltage greater than 10.4 volts.

Set Condition: No sensor output is received when the PCM powers up the sensor heater. O2 heater is out of control for 128 seconds after it has reached 300 deg C. Two trip fault.

P0161-O2 SENSOR 2/2 HEATER PERFORMANCE

When Monitored: Engine running and heater duty cycle greater than 0%. ASD Relay is energized. Battery voltage greater than 10.4 volts.

Set Condition: No sensor output is received when the PCM powers up the sensor heater. O2 heater is out of control for 128 seconds after it has reached 350 deg C. Two trip fault.

POSSIBLE CAUSES
O2 SENSOR HEATER OPERATION
O2 HEATER ELEMENT
O2 HEATER CONTROL CIRCUIT OPEN
O2 HEATER GROUND CIRCUIT OPEN